

Final Project Report Template

1. Introduction

1.1. Project overviews

The **Food Ordering Behaviour and Consumer Trends** project is designed to analyze customer food ordering patterns and restaurant performance using Tableau. The project focuses on transforming raw restaurant order data into meaningful business insights through data preparation, visualization, interactive dashboards, and Tableau Stories. The analysis includes customer preferences across different cuisines, meal types, restaurant types, customer ratings, city-wise order distribution, repeat ordering behaviour, and average order value. Interactive filters, parameters, and Key Performance Indicators (KPIs) are incorporated to enable users to explore the data dynamically and make informed decisions.

The project demonstrates the complete business intelligence workflow, starting from data collection and preprocessing to visualization development, dashboard design, story creation, and project documentation. The insights generated can help restaurant owners and food delivery businesses understand customer behaviour, identify market trends, improve customer satisfaction, and support data-driven business decision-making.

1.2. Objectives

- To analyze customer food ordering behaviour using Tableau.
- To identify the most popular cuisines and meal types.
- To compare restaurant performance based on order volume and average order value.
- To analyze customer ratings across different meal types.
- To identify the top-performing cities based on food orders.
- To study repeat ordering behaviour.
- To develop an interactive dashboard and Tableau Story for business insights.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The project focuses on analyzing food ordering behaviour and consumer trends from the perspective of different stakeholders. Restaurant owners and food delivery businesses generate large amounts of order-related data, but understanding customer preferences,

popular cuisines, meal types, ratings, revenue trends, and city-wise ordering patterns from this data can be difficult when analyzed manually.

The identified customer problem statements are:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A customer who orders food online.	Choose food that matches my preferences, meal type, hunger level, mood, and budget.	There are many food choices and it is difficult to understand which options are most suitable.	Ordering decisions can be influenced by cuisine, discounts, restaurant type, location, weather, and other customer factors.	Uncertain about my choices and whether I am getting good value from my order.
PS-2	A food-ordering business or restaurant manager.	Understand customer preferences and increase orders, order value, and repeat purchases.	Customer <u>behavior</u> and preferences are spread across many different factors and are difficult to compare manually	there are patterns involving cuisine, city, meal type, discounts, repeat orders, ratings, mood, hunger, and weather.	it is difficult to identify the most important customer segments and make data-driven decisions.

Therefore, the main problem is the **difficulty in converting large food-ordering datasets into meaningful and easily understandable business insights**. A structured data analysis and visualization approach is required to help stakeholders understand consumer behaviour and make better business decisions.

2.2. Project Proposal (Proposed Solution)

Project Overview

The proposed project, **Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits**, aims to analyze food ordering behaviour and consumer trends using **Tableau**.

Objective

The objective of the project is to analyze food ordering behaviour using Tableau by identifying popular cuisines, customer preferences, meal types, ratings, and city-wise ordering patterns. The project aims to provide interactive visual insights that support better business decision-making.

Scope

The project focuses on analyzing restaurant and food-ordering data using Tableau. It includes data preparation, visualization, dashboard development, and story creation to identify customer behaviour and restaurant performance trends. The project is limited to the provided food-ordering dataset.

Problem Statement

Restaurant owners and food delivery businesses generate large amounts of order data. Analyzing this data manually is difficult, making it challenging to understand customer preferences, popular cuisines, revenue trends, and overall business performance.

Impact

The proposed solution helps businesses make data-driven decisions by providing interactive dashboards and visual insights. It supports menu planning, marketing strategies, customer satisfaction, and operational improvements.

Proposed Solution

The proposed solution is to use **Tableau** to connect, clean, analyze, and visualize the foodordering dataset. Interactive charts, filters, KPIs, parameters, dashboards, and stories are developed to understand customer ordering behaviour and restaurant performance effectively.

Approach

The project uses Tableau to connect, clean, and analyze the restaurant dataset. Various charts, filters, KPIs, parameters, dashboards, and stories are created to visualize customer ordering behavior and restaurant performance effectively.

Key Features

- Interactive Tableau Dashboard
- Customer Age Distribution Analysis
- Meal Type Analysis
- Top 5 Cities by Order Volume
- Weekday vs Weekend Analysis
- Discount Analysis
- Customer Mood Analysis
- City vs Cuisine Analysis
- Average Order Analysis by Hunger Level
- Customer Rating and Preference Analysis
- Tableau Story for Consumer Behaviour Insights

Resource Requirements Hardware

Requirements:

- Intel Core i3 / Ryzen 3 or higher
- Dual-Core Processor
- 8 GB RAM
- Minimum 10 GB free storage space

Software Requirements:

- Tableau Desktop as the development environment

Data Requirements:

- Food Ordering Dataset
- CSV format

2.3. Initial Project Planning

The project was planned using a sprint-based approach. The work was divided into functional requirements and user stories so that data collection, loading, preparation, visualization, dashboard/story development, documentation, and demonstration could be completed systematically.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Dataset Collection	USN-1	As a data analyst, I want to download the restaurant dataset so that I can start the project.	2	High	Suyashri Satappa Patil	9-08-2026	9-08-2026
Sprint-1	Data Loading	USN-2	As a data analyst, I want to load the dataset into Tableau for analysis.	1	High	Suyashri Satappa Patil	10-08-2026	10-08-2026
Sprint-1	Data Preparation	USN-3	As a data analyst, I can verify and prepare the dataset fields in Tableau so that dimensions and measures are correctly identified.	3	Low	Suyashri Satappa Patil	15-08-2026	16-08-2026
Sprint-1	Data Visualization	USN-4	As a data analyst, I can create sheets for Orders by City, Age Distribution, and Meal Type so that I can understand customer ordering patterns.	5	Medium	Suyashri Satappa Patil	16-08-2026	19-08-2026
Sprint-2	Dashboard & Story Development	USN-5	As a business user, I want an interactive dashboard and story so that I can easily understand business insights.	5	High	Suyashri Satappa Patil	21-08-2026	23-08-2026
Sprint-2	Publishing, Documentation & Demo Recording	USN-6	As a dashboard user, I can view the completed Tableau dashboard and story so that the analysis can be presented and shared with stakeholders.	8		Suyashri Satappa Patil	24-08-2026	26-08-2026

The sprint plan provides a structured sequence for completing the project from dataset collection through visualization, dashboard and story development, documentation, and final demonstration.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection Plan

The dataset was downloaded from the **SkillWallet Project Workspace** after logging into the SkillWallet portal. The downloaded dataset was then imported into **Tableau Desktop** for data preparation, visualization, dashboard development, and story creation.

Raw Data Sources Identified

The project uses a restaurant and food-ordering dataset downloaded from the SkillWallet Project Workspace. The dataset contains information related to:

- Customer orders
- Cuisines
- Meal types
- Customer ratings
- Restaurant types
- City details
- Revenue
- Delivery information

This information is used to analyze customer behaviour, ordering patterns, restaurant performance, and other business-related insights.

Source Name	Description	Location /URL	Format	Size	Access Permissions
food_ordering_ behavior_dataset	Contains customer orders, cuisines, meal types, customer ratings, restaurant types, city details, revenue, and delivery information used for business analysis.	SkillWallet Project Workspace (Downloaded Dataset)	CSV	5.07 MB	SkillWallet Project Access

3.2. Data Quality Report

The Data Quality Report was prepared to identify and assess potential data quality issues in the selected dataset before performing analysis and visualization. The dataset was reviewed for missing values, duplicate records, incorrect data types, and inconsistent categorical values.

Data Source	Data Quality Issue	Severity	Resolution Plan
food_ordering_behavior_dataset	The dataset was checked for missing values, duplicate records, incorrect data types, and inconsistent categorical values before analysis. No major data quality issues affecting the analysis were identified.	Low	The dataset was validated by assigning appropriate data types in Tableau, verifying data consistency, and using the cleaned dataset for visualization, dashboard development, and story creation.

Data Quality Assessment

The dataset was reviewed before analysis to ensure that the available data was suitable for visualization and business analysis. The assessment focused on:

- Missing values
- Duplicate records
- Incorrect data types
- Inconsistent categorical values

No major data quality issues affecting the analysis were identified. Appropriate data types were assigned in Tableau, and data consistency was verified before using the dataset for visualization, dashboard development, and story creation.

3.3. Data Exploration and Preprocessing

The data exploration and preprocessing phase was performed to understand the dataset, identify potential data-quality issues, and prepare the data for reliable analysis and visualization in Tableau.

Data Overview

The project uses the **food_ordering_behavior_dataset** downloaded from the SkillWallet Project Workspace. The dataset is available in **CSV format** with a size of **5.07 MB**. It contains information such as customer orders, cuisines, meal types, customer ratings,

restaurant types, cities, revenue, delivery time, and other attributes used for restaurant analytics.

Data Cleaning

The dataset was reviewed to identify missing values, duplicate records, incorrect data types, and inconsistent categorical values. Data consistency was verified, and the dataset was prepared for analysis in Tableau to ensure reliable visualizations and business insights.

Data Transformation

The dataset was transformed in Tableau by applying filters, sorting data, grouping relevant fields, and creating calculated fields where required for visualization and dashboard analysis.

Data Type Conversion

Appropriate data types such as **String, Number, and Date** were verified and assigned to the dataset fields in Tableau to ensure accurate analysis and calculations.

Column Splitting and Merging

No column splitting or merging was required because the dataset structure was already suitable for analysis in Tableau.

Data Modeling

The project used a **single dataset**; therefore, no table relationships or joins were required. Measures and dimensions available in the dataset were used to create visualizations, dashboards, and stories in Tableau.

Save Processed Data

The cleaned and prepared dataset was saved within the **Tableau workbook** and used to create interactive visualizations, dashboards, and stories for business analysis.

4. Data Visualization

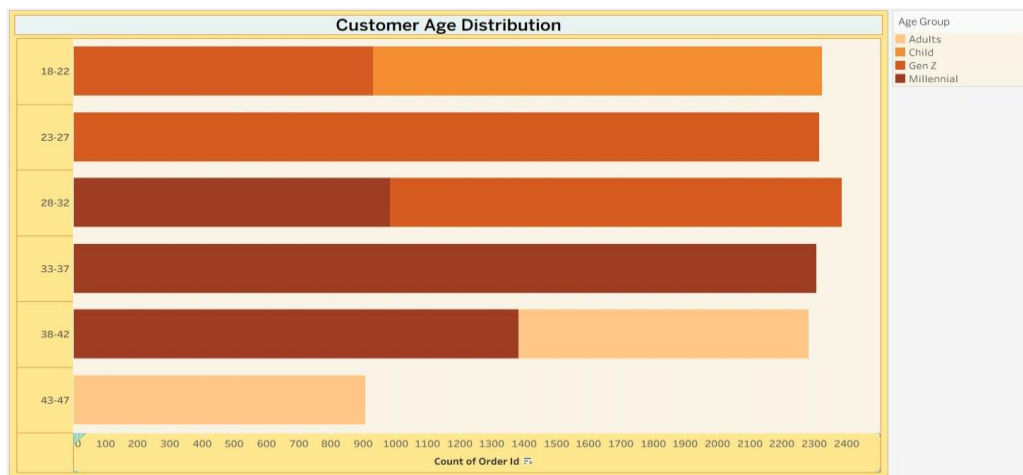
4.1. Framing Business Questions

- 1) Which age groups contribute the highest number of customers?
- 2) Which meal type receives the highest number of orders?
- 3) Which are the top 5 cities by order volume?
- 4) How does food ordering behaviour differ between weekdays and weekends?
- 5) What is the distribution of customer mood across food orders?
- 6) How does hunger level affect the average order value?

- 7) Which cuisines are most popular in different cities?
- 5) How do discounts influence customer ordering behaviour?

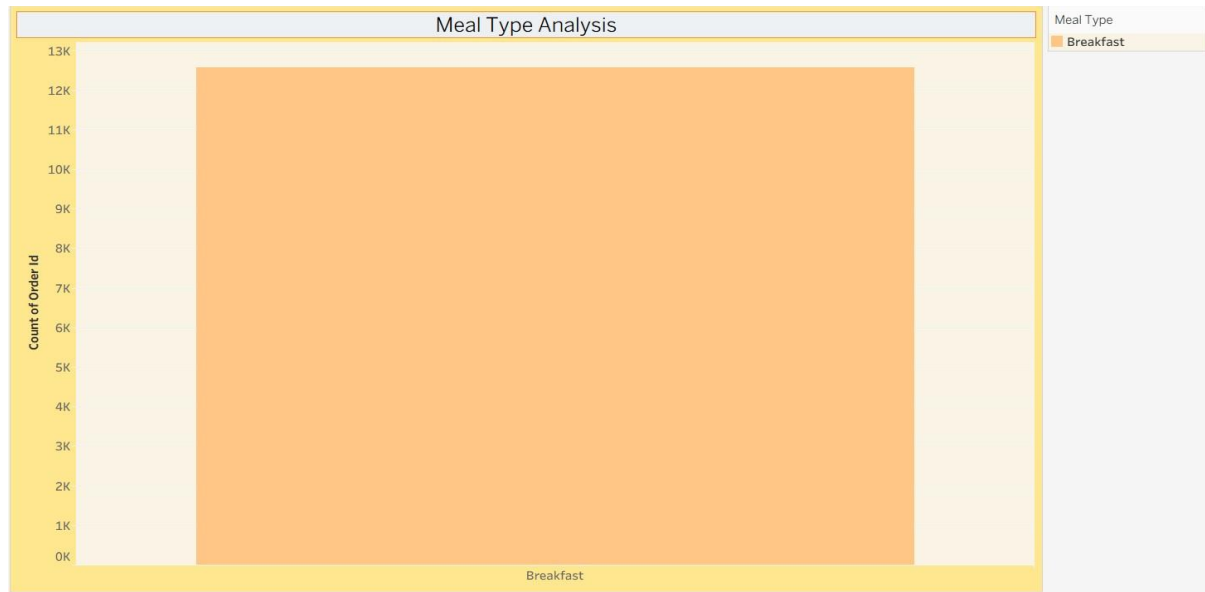
4.2. Developing Visualizations

- **Business Question 1 .**
- **Visualization:** Bar chart showing the distribution of customers across different age groups.
- **Screenshot of Visualization:**

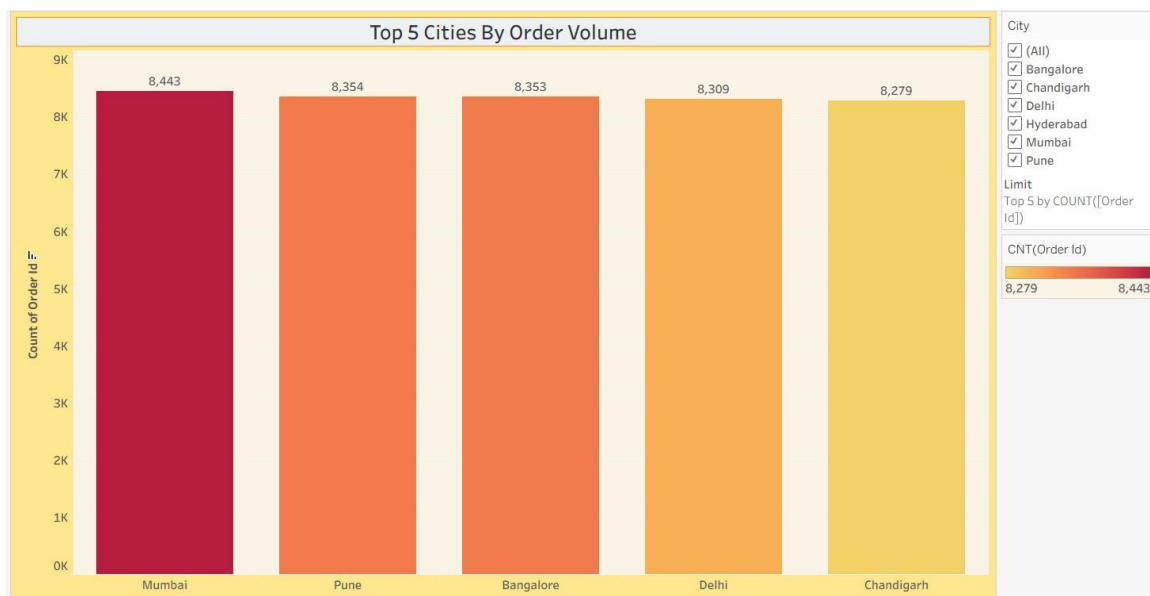


Business Question 2 ○ **Visualization:** Bar chart comparing the total number of orders across different meal types such as breakfast, lunch, dinner, and snacks.

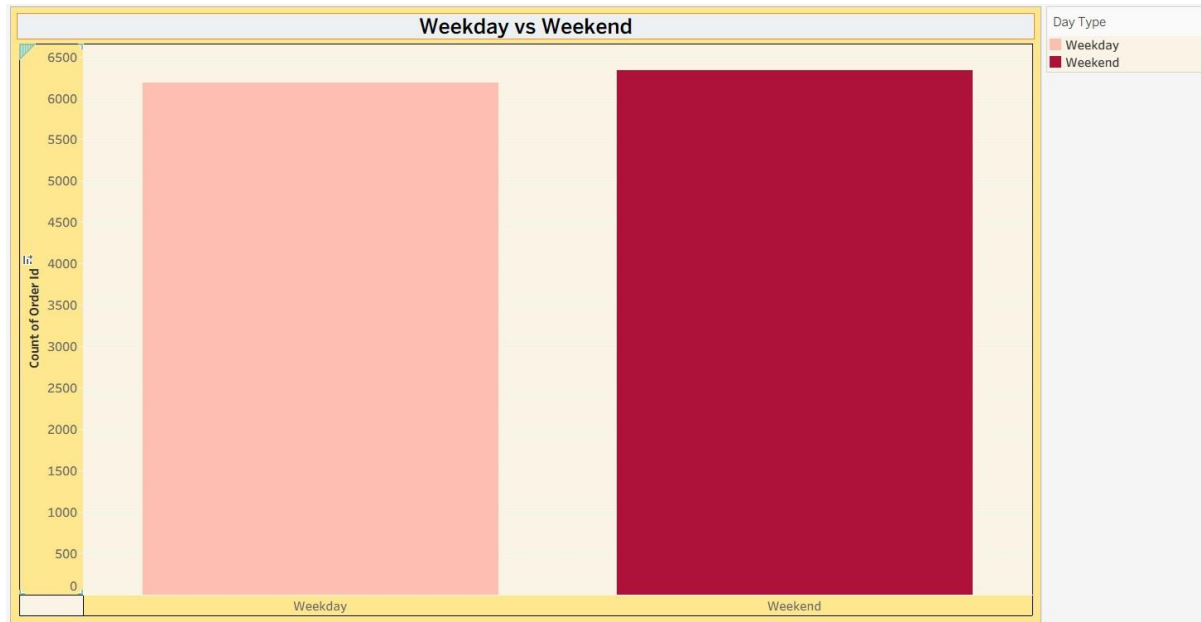
- **Screenshot of Visualization:**



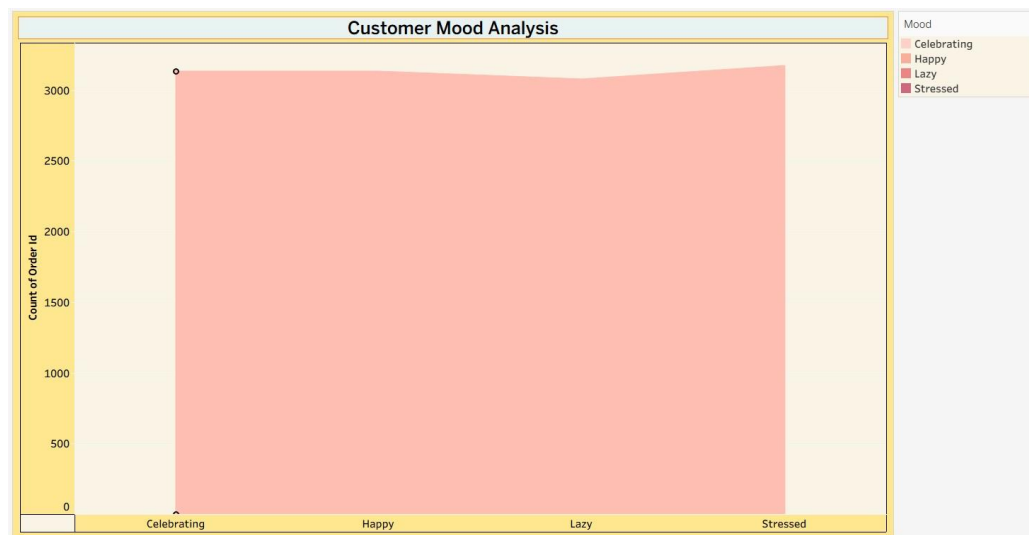
Business Question 3 ○ **Visualization: Visualization:** Horizontal bar chart showing the top five cities based on total food order volume. ○ **Screenshot of Visualization:**



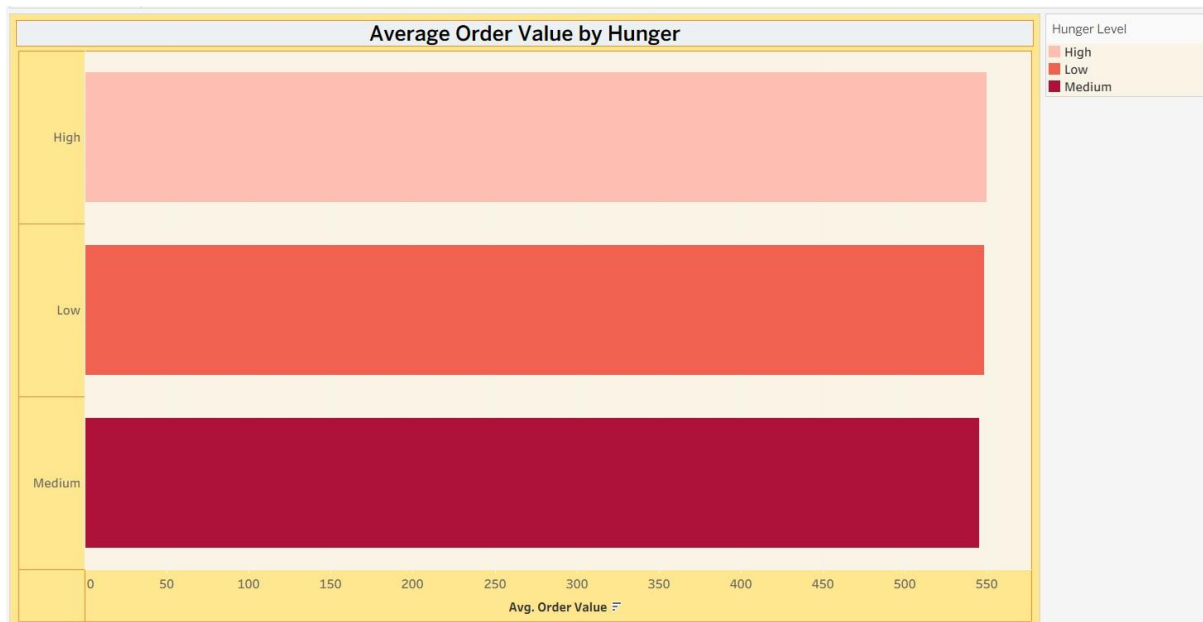
Business Question 4 ○ **Visualization:** Bar chart comparing total orders on weekdays and weekends. ○ **Screenshot of Visualization:**



Business Question 5 ○ **Visualization:** Bar chart showing the distribution of orders across different customer mood categories. ○ **Screenshot of Visualization:**

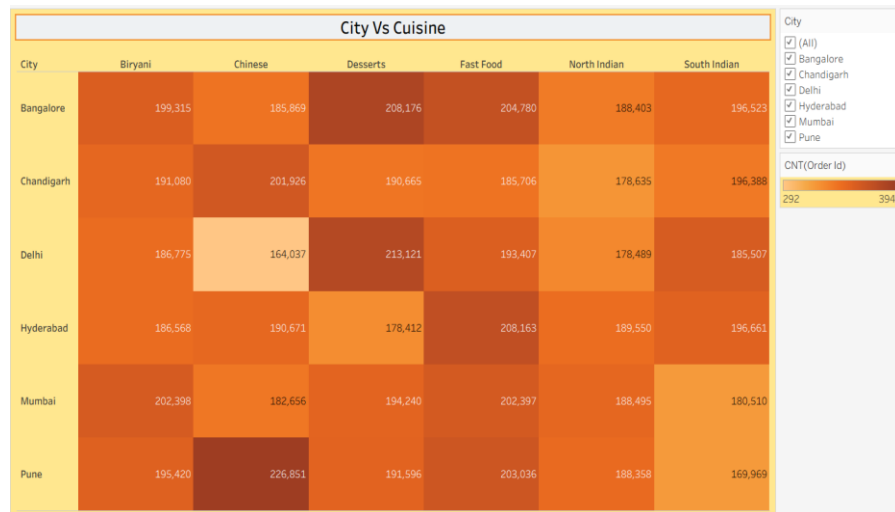


Business Question 6 ○ **Visualization:** Bar chart showing the average order value for different hunger levels. ○ **Screenshot of Visualization:**

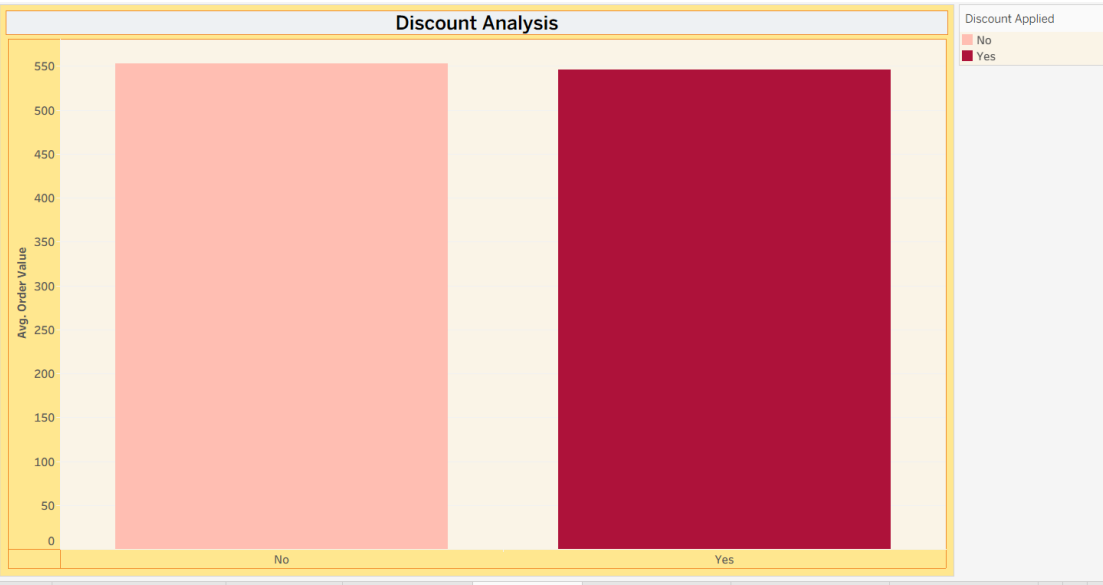


Business Question 7 ○ **Visualization:** Bar chart comparing orders with and without discounts.

○ **Screenshot of Visualization:**

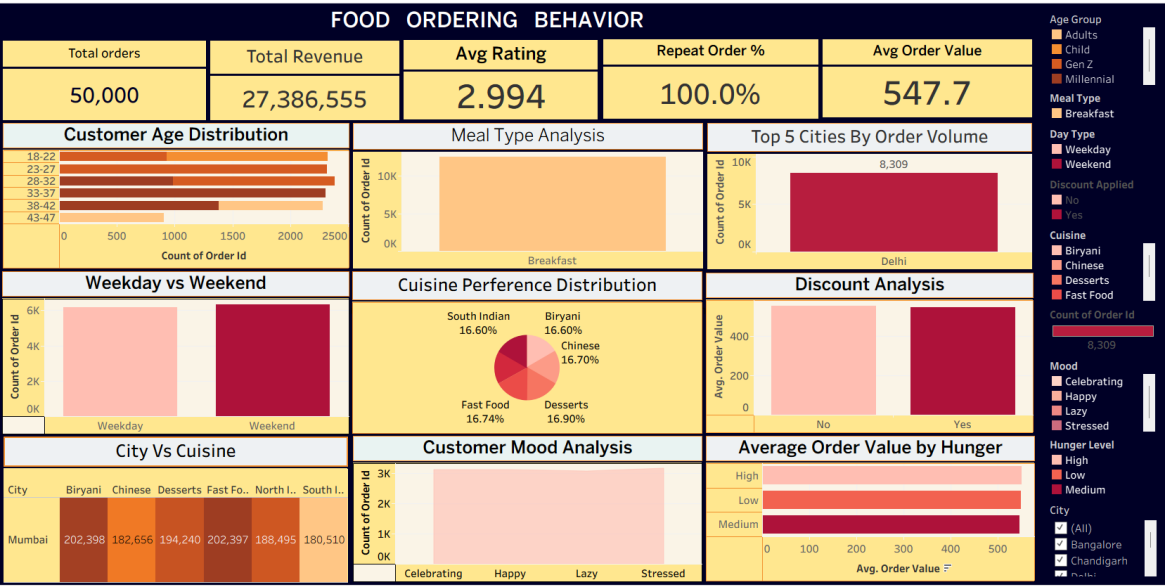


Business Question 8 ○ **Visualization:** Bar chart comparing orders with and without discounts. ○ **Screenshot of Visualization:**



5. Dashboard

5.1. Dashboard Design File



Major Outcomes

- Identified the most preferred cuisines, helping understand customer food preferences and popular menu categories.
 - Identified the leading cities by order volume, highlighting the strongest markets and locations with higher customer demand.
 - Analyzed customer age distribution, helping identify the major customer segments using food-ordering services.
 - Determined popular meal types, providing insights into customers' preferred ordering periods.
 - Compared weekday and weekend ordering behaviour, revealing differences in customer ordering patterns.
 - Analyzed the impact of discounts on orders, helping understand how promotional offers influence customer purchasing behaviour.
 - Analyzed customer mood and satisfaction patterns, providing insights into the overall customer experience.
- Compared cities and cuisines, identifying differences in cuisine preferences across locations.
- Examined average order value by hunger level, helping understand the relationship between hunger and customer spending.
 - Developed an interactive Tableau dashboard and story, combining multiple analyses into a single visual solution for easier interpretation and data-driven decision-making.

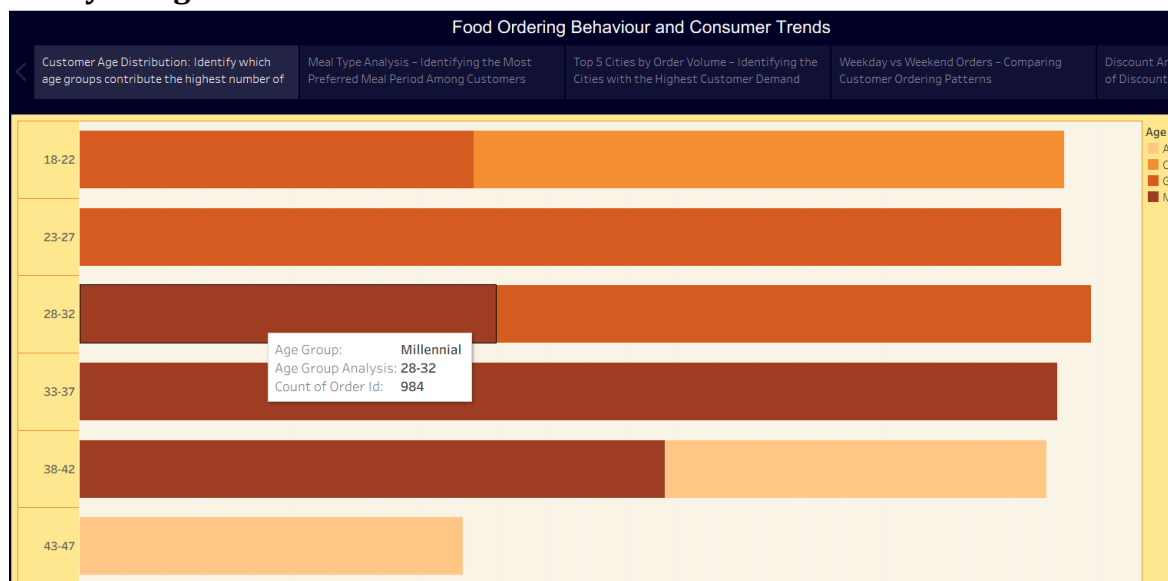
Activity 2

Dashboard Public Link:

https://public.tableau.com/views/Food_ordering_dashboard/Dashboard1?:language=enUS&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

6. Report

6.1. Story Design File



Observations

- Young adult customers represented a significant share of the customer base, indicating that this age group is an important target segment for food-ordering businesses.
- Meal type analysis showed differences in order volume across breakfast, lunch, dinner, and snacks, helping identify customers' preferred ordering periods.
- Mumbai recorded the highest order volume among the top five cities, highlighting strong customer demand and making it an important market for food-ordering businesses.

- Cuisine preferences varied across different cities, showing that customer food choices are influenced by location and local demand.
- Weekday and weekend ordering patterns showed differences in customer behaviour, providing useful insights into when customers are more likely to place food orders.
- Discount usage influenced customer ordering behaviour, helping identify the role of promotional offers in attracting customers and encouraging orders.
- Customer mood analysis provided insights into the overall customer experience, helping businesses understand how ordering experiences may relate to customer satisfaction.
- Average order value varied according to hunger level, indicating that customers' hunger conditions can influence their spending and order size.
- The Tableau Story combines customer demographics, meal preferences, city trends, cuisine choices, discounts, mood, and hunger-level analysis, providing a step-by-step view of food ordering behaviour and supporting data-driven business decisions.

• **Activity 2:**
Story Public Link:

https://public.tableau.com/views/Food_ordering_behavior_analysis/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing

7.1 Utilization of Data filters

Sr. No.	Filter / Parameter Used	Purpose
1	Cuisine Parameter	Analyze data for the selected cuisine.
2	City Parameter	View customer orders by city.
3	Meal Type Parameter	Analyze orders based on meal type.

7.2 No of Calculation Field

Sr. No.	Calculated Field	Purpose
1	Age Group	Categorizes customers into different age groups for analysis.

7.3 No of Visualization

Sr. No.	Visualization Name
1	City Vs Cuisine
2	Orders By Age Groups And Restaurant Type
3	Top 5 Cities By Order Volume
4	Meal Type Analysis
5	Weekday vs Weekend
6	Discount Analysis
7	Customer Mood Analysis
8	Customer Age Distribution
9	Average Order Value by Hunger

8. Conclusion/Observation

The **Food Ordering Behaviour and Consumer Trends** project successfully analyzed customer ordering patterns using Tableau. The project transformed raw data into meaningful insights through interactive visualizations, dashboards, and stories. It identified popular cuisines, restaurant performance, city-wise order distribution, customer ratings, repeat ordering behaviour, and average order values. The developed dashboard enables users to explore the data interactively and supports data-driven decision-making for restaurants and food delivery businesses.

9. Future Scope

- Integrate real-time food ordering data for continuous analysis.
- Apply predictive analytics to forecast customer demand and ordering trends.
- Develop personalized food recommendation systems based on customer preferences.
- Expand the dashboard with additional KPIs and advanced business analytics.
- Integrate the dashboard with web or mobile applications for broader accessibility.

10. Appendix

10.1. Source Code(Calculated Field)

Calculated Field Name: Age Group

IF [Age] <= 20 THEN "Child"

ELSEIF [Age] <= 30 THEN "Gen Z"

ELSEIF [Age] <= 40 THEN "Millennial"

ELSE "Adults"

END

10.2. GitHub & Project Demo Link GitHub

Repository:

<https://github.com/SuyashriPatil/Food-Ordering-Behaviour-and-Consumer-Trends->

Project Demo Video:

<https://drive.google.com/file/d/1u9DJcwAGGerI78NKEJXevr0x4bLcduep/view?usp=sharing>